BI-FUEL F-SERIES CATALYTIC CONVERTER REPLACEMENT

AFFECTED VEHICLES: **CERTAIN 2001 MODEL YEAR F-SERIES EQUIPPED WITH 5.4L BI-FUEL ENGINE**

OVERVIEW

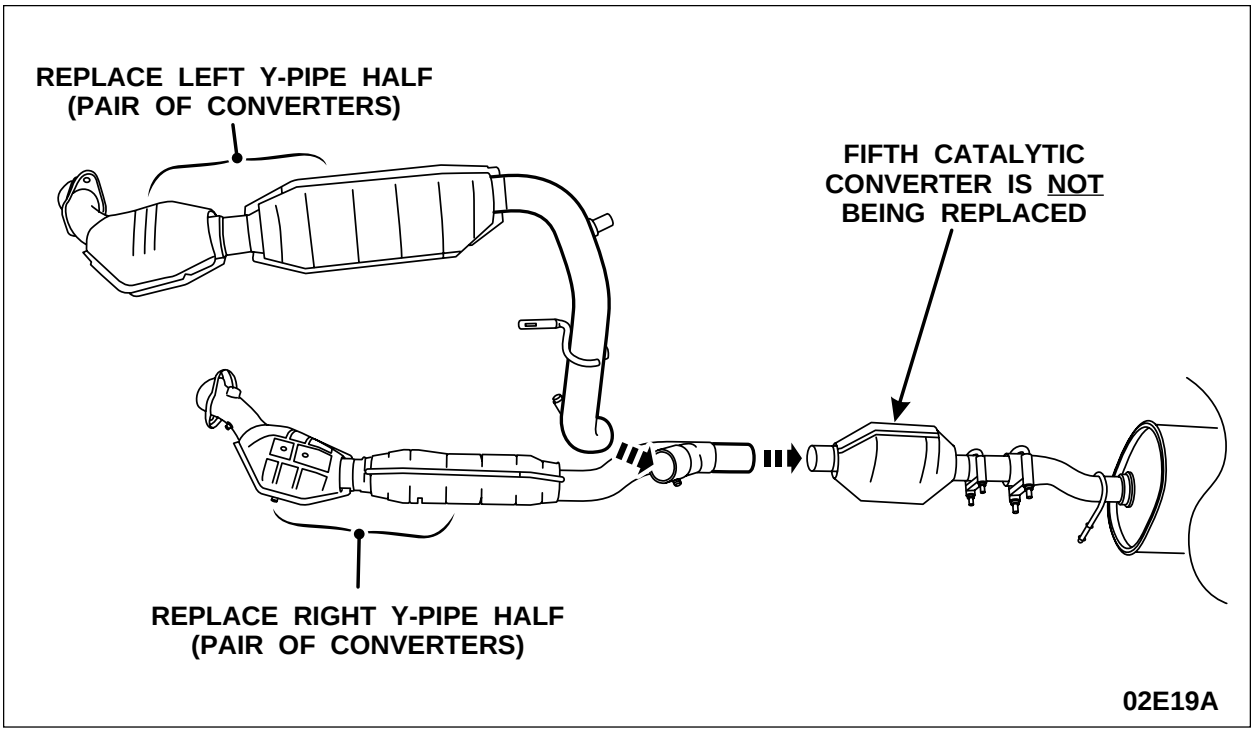
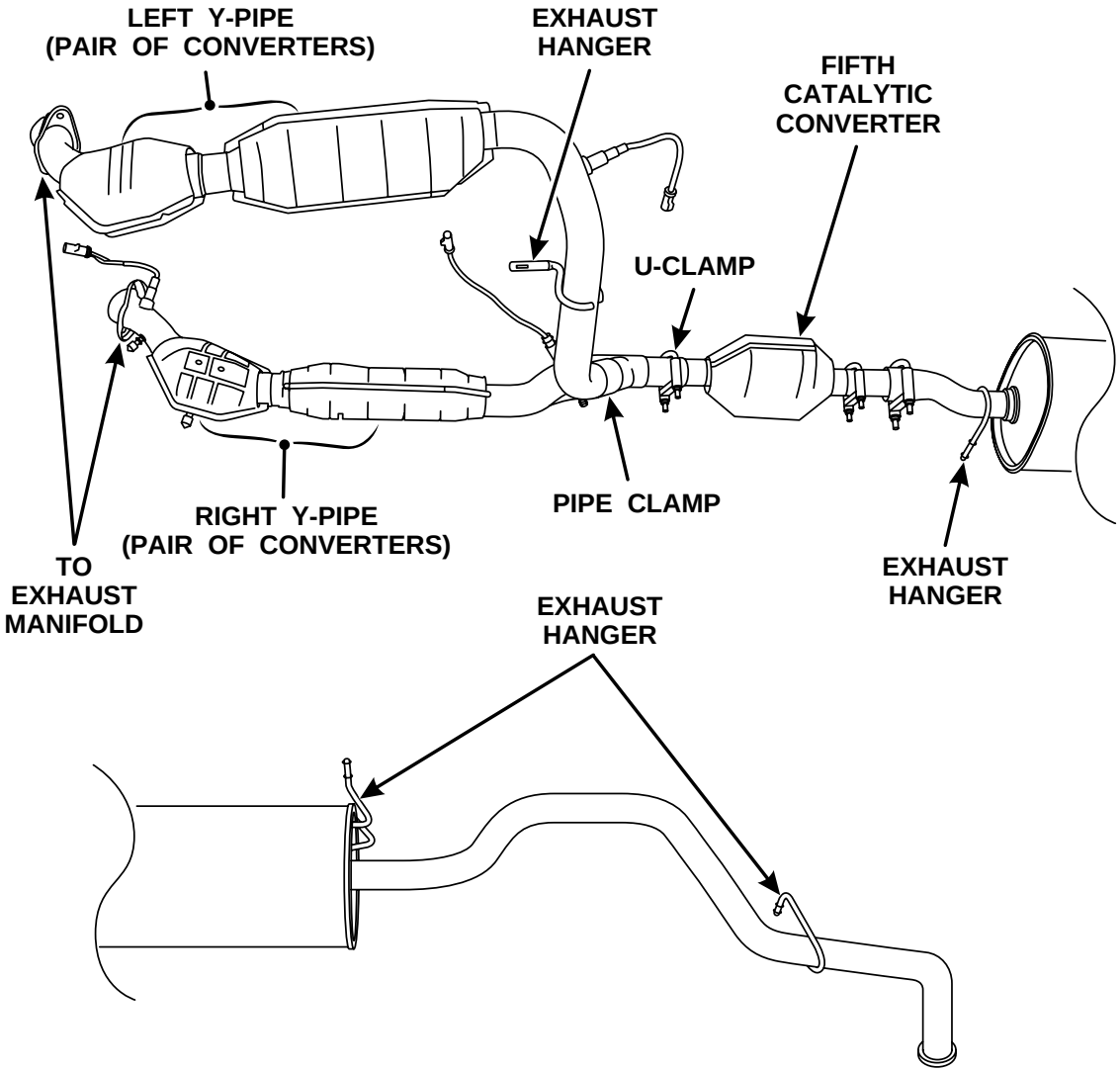


FIGURE 1

The affected vehicles are equipped with five (5) catalytic converters. The left and right halves of the Y-pipe are assemblies that include a pair of catalytic converters each. This procedure is for the replacement of both the left and right Y-pipe halves (total of four catalytic converters). The fifth catalytic converter is located between the Y-pipe assembly outlet and the muffler. The fifth catalytic converter is not being replaced. Therefore, extreme care must be taken not to damage the fifth catalytic converter when separating it from the Y-pipe. Also, an authorized modifications label must be completed and installed on the vehicle.

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EXHAUST COMPONENT LOCATION

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FIGURE 2

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REMOVAL

NOTE: Refer to Figures 1 and 2 in the Overview section for exhaust component location.

1. Raise the vehicle on a hoist.
2. On 4x4 vehicles, remove the transfer case skid plate.
3. Remove the U-clamp from the inlet pipe of the fifth (rearmost) catalytic converter.
4. Starting at the rear of the vehicle, disconnect the three (3) rubber exhaust insulators from the exhaust hangers.
5. Separate the fifth catalytic converter inlet pipe from the front Y-pipe and lower the converter/muffler/exhaust pipe assembly. If the pipes will not separate, saw the Y-pipe off in front of the fifth catalytic converter inlet pipe. Then remove the section of the Y-pipe from inside the fifth catalytic converter inlet pipe, using standard exhaust repair techniques.
6. Disconnect the four (4) heated exhaust gas oxygen (HEGO) sensors.
7. Loosen, but do not remove, all four (4) catalytic converter nuts from the exhaust manifolds.
8. Remove the left and right heat shield bolts from the transmission crossmember.
9. Support the transmission with a suitable transmission jack.
10. Remove the two (2) transmission mount nuts at the crossmember.
11. Reposition the wiring harness from the outboard side of the left frame rail to allow removal of the crossmember stud plate.
12. Remove the stud plates and crossmember.
13. Remove the exhaust hanger bolt at the transmission mount.
14. Loosen the clamp that connects the left and right catalytic converters.
15. Remove the nuts from the left catalytic converter at the exhaust manifold.
16. Separate and remove the left catalytic converter from the right catalytic converter.
17. Remove the nuts from the right catalytic converter at the exhaust manifold, then remove the right catalytic converter.

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INSTALLATION

1. One at a time, transfer the HEGO sensors from the original catalytic converters into the ***new*** catalytic converters. Tighten the sensors to 40 Nm (30 lb-ft).
2. With the aid of an assistant, install the right catalytic converter. Install ***new*** nuts at the exhaust manifold, but do not tighten them at this time.
3. With the aid of an assistant, slide the left catalytic converter into the right catalytic converter, but do not tighten the clamp at this time. Install ***new*** nuts at the exhaust manifold, but do not tighten them at this time.
4. Install the exhaust hanger to the transmission mount with a ***new*** bolt, but do not tighten at this time.
5. Install the crossmember with the stud plates and nuts. Tighten the nuts to 90 Nm (66 lb-ft).
6. Remove the transmission jack.
7. Reposition the wiring harness on the left frame rail.
8. Install the two (2) transmission mount nuts. Tighten the nuts to 103 Nm (76 lb-ft).
9. Install the left and right heat shield bolts.
10. Connect the four (4) HEGO sensors.
11. If required, straighten the inlet pipe of the fifth catalytic converter with a commercially available tailpipe expander tool.
12. Slide the converter/muffler/exhaust pipe onto the Y-pipe but do not tighten the clamp at this time.
13. Connect the three (3) rubber exhaust insulators to the exhaust hangers.
14. Starting at the front of the vehicle, align and adjust the complete exhaust system, checking clearances between the exhaust and all frame and body components. Tighten the exhaust components in the following order:
    - Right and left catalytic converter-to-exhaust manifold nuts to 40 Nm (30 lb-ft).
    - Right and left catalytic converter pipe clamp to 48 Nm (35 lb-ft).
    - Exhaust hanger-to-transmission mount bolt to 40 Nm (30 lb-ft).
    - Fifth catalytic converter inlet pipe U-clamp to 48 Nm (35 lb-ft).
15. On 4x4 vehicles, install the transfer case skid plate.
16. Lower the vehicle.
17. Start the engine and check for exhaust leaks.

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PREPARE AUTHORIZED MODIFICATIONS LABEL AND PROVIDE PROOF OF CORRECTION CERTIFICATE

1. Obtain and prepare the Authorized Modifications label. Write the modification completed, your P&A code, date of repair and the recall program number in the spaces provided on the label. See Figure 3.

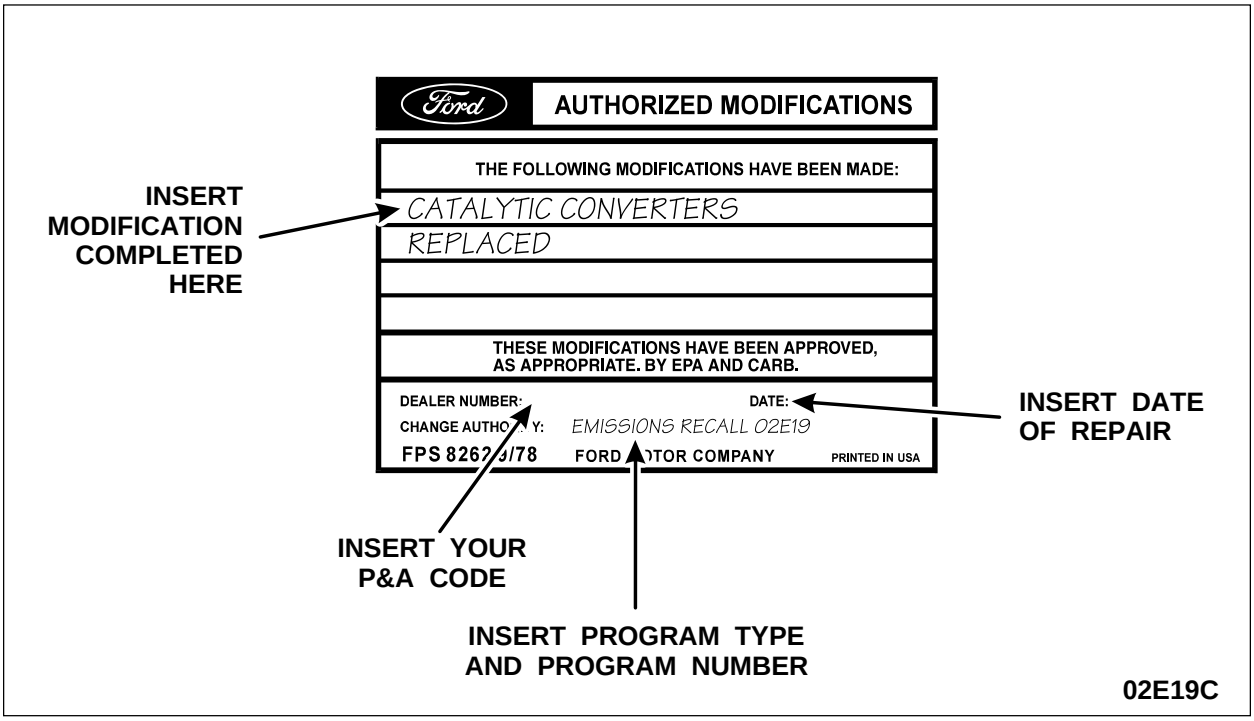


FIGURE 3

2. Clean an area of the radiator support with isopropyl alcohol and a clean cloth.
3. Apply the label to the clean area.
4. Provide California owners with a Vehicle Emission Recall–Proof of Correction certificate.

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